AUROADAS: A Modular Multi-Sensor Fusion and Perception System for Long-Sighted ADAS→ADS Autonomy

Abstract

Advanced Driver Assistance Systems (ADAS) are evolving toward higher-order Autonomous Driving Systems (ADS), requiring perception and localization pipelines that remain robust under uncertainty, degraded sensing, and environmental complexity. AUROADAS introduces a modular, multi-sensor fusion architecture that integrates stereo vision, LiDAR, IMU, GPS, and wheel odometry into a unified factor-graph framework. The system provides stable localization, semantic perception continuity, and real-time performance on embedded hardware. AUROADAS is evaluated across KITTI [1], EuRoC [2], and CARLA [3], demonstrating sub-percent drift, high tracking retention, and resilience under sensor dropouts. The system incorporates ADS-level safety mechanisms aligned with ISO 21448 SOTIF [4], including redundancy, uncertainty modeling, graceful degradation, and transparent verification. Results show that AUROADAS provides a scalable autonomy backbone suitable for long-sighted ADAS→ADS deployment.

Index Terms

Autonomous Driving Systems, ADAS, Multi-Sensor Fusion, Factor Graphs, Perception, Localization, ROS2, Embedded Systems, SOTIF, Safety Case.

I. Introduction

Autonomous driving systems require perception and localization pipelines capable of maintaining situational awareness under uncertainty, degraded sensing, and dynamic environments. Traditional ADAS systems rely primarily on single-modality sensing and deterministic pipelines, which limits robustness in complex scenarios. In contrast, ADS autonomy demands multi-modal redundancy, probabilistic modeling, and predictable behavior under environmental complexity. Recent advances in factor-graph-based fusion [5], visual-inertial odometry [6], and LiDAR-based mapping [7] highlight the importance of tightly coupled multi-sensor architectures for reliable autonomy.

AUROADAS addresses these challenges by integrating stereo vision, LiDAR, IMU, GPS, and wheel odometry into a unified factor-graph architecture capable of stable localization under degraded sensing. The system architecture is shown in **Fig. 1**, illustrating the layered design from sensor acquisition to perception and mapping. A ROS-based software architecture (**Fig. 2**) enables modular deployment, real-time execution, and integration with OEM drive-by-wire systems.

The transition from ADAS to ADS introduces several challenges. Perception systems must maintain object identity and semantic consistency under occlusion, motion blur, and dynamic motion. Localization systems must remain stable under sensor degradation, including camera blackout, GPS dropout, and LiDAR sparsity. The autonomy stack must incorporate uncertainty modeling and fallback behavior aligned with ADS safety requirements. Furthermore, the system must operate within real-time constraints on embedded platforms such as the NVIDIA Jetson Orin AGX. AUROADAS is designed to address these challenges through a modular architecture that scales from ADAS-level assistance to ADS-level autonomy.

The AUROADAS pipeline is evaluated across KITTI [1], EuRoC [2], and CARLA [3], and validated under controlled degradation scenarios including sensor dropouts and environmental perturbations. The system demonstrates robust performance aligned with ADS safety expectations, including redundancy, uncertainty modeling, graceful degradation, and transparent verification.

A. Contributions

This work provides the following contributions:

1. **Unified multi-sensor fusion architecture***:* A factor-graph framework integrating stereo vision, LiDAR, IMU, GPS, and wheel odometry, enabling stable localization under degraded sensing conditions.
2. **Modular ROS2-based autonomy pipeline***:* A scalable architecture supporting ADAS-level lane keeping and object detection, and ADS-level global mapping, loop closure, and semantic scene understanding.
3. **Robust perception stack***:* A detection-tracking-semantics pipeline that maintains object identity and semantic continuity under occlusion and dynamic motion.
4. **Embedded deployment:** Real-time performance (22–28 ms latency) on the Jetson Orin AGX, demonstrating suitability for production-grade embedded autonomy.
5. **Comprehensive ADS safety case***:* A complete ISO 21448 SOTIF-aligned safety case including hazard analysis, FMEA, requirements traceability, ODD specification, and safety validation plan.
6. **Extensive experimental evaluation:** Demonstrated performance across KITTI [1], EuRoC [2], CARLA [3], sensor dropout scenarios, and ablation studies, achieving sub-percent

II. RELATED WORK

Research in autonomous driving has produced a wide range of perception and localization systems, each addressing different aspects of robustness, environmental complexity, and multi-modal fusion. Classical visual SLAM systems such as ORB-SLAM2 [8] and DSO [9] demonstrated strong performance in feature-rich environments but remain sensitive to illumination changes, motion blur, and texture sparsity. Visual-inertial odometry approaches, including OKVIS [10] and VINS-Mono [11], improved robustness by incorporating inertial constraints, yet still rely heavily on stable visual tracking. LiDAR-centric systems such as LOAM [7] and its variants provide strong geometric consistency but lack semantic richness and struggle in environments with sparse or reflective surfaces.

Multi-sensor fusion has emerged as a critical direction for ADS-level autonomy. Factor-graph-based approaches such as GTSAM [5] and iSAM2 [12] provide a principled framework for integrating heterogeneous measurements, enabling real-time optimization and uncertainty modeling. Recent works have explored tightly coupled visual-LiDAR-IMU fusion [13], GPS-aided SLAM [14], and odometry-enhanced mapping [15], demonstrating improved resilience under degraded sensing. AUROADAS builds on these foundations by integrating stereo vision, LiDAR, IMU, GPS, and wheel odometry into a unified factor-graph backend capable of stable operation under sensor dropouts and environmental perturbations.

Perception research has advanced significantly with deep learning-based object detectors [16], multi-object trackers [17], and semantic mapping frameworks [18]. However, ADS-level perception requires not only accurate detection but also temporal consistency, identity retention, and alignment with the ego-pose. AUROADAS incorporates a perception stack that leverages fused localization to stabilize detections and maintain semantic continuity, addressing challenges observed in prior ADAS-focused systems.

Safety frameworks such as ISO 21448 SOTIF [4] and UL 4600 [19] emphasize the importance of uncertainty modeling, redundancy, and verifiable behavior in autonomous systems. Recent studies highlight the need for safety-aligned perception [20], robust fallback behavior [21], and transparent fusion architectures [22]. AUROADAS contributes to this direction by providing a complete ADS safety case, including hazard analysis, FMEA, requirements traceability, ODD specification, and validation plan.

In summary, AUROADAS advances the state of the art by combining multi-modal factor-graph fusion, robust perception, embedded deployment, and ADS-aligned safety mechanisms into a unified architecture. The system architecture is illustrated in **Fig. 1**, and the ROS-based software pipeline is shown in **Fig. 2**, providing a foundation for the methodology described in the following sections.

III. SYSTEM ARCHITECTURE

The AUROADAS system architecture is designed as a modular, ROS2-based autonomy pipeline that scales from ADAS-level assistance to ADS-level autonomous operation. The architecture is organized into five layers: the Sensor Interface Layer, Visual Front-End, Multi-Sensor Fusion Layer, Mapping Layer, and Perception Layer. Each layer is implemented as a ROS2 node or node group, enabling distributed execution, real-time communication, and integration with OEM drive-by-wire systems. The overall architecture is illustrated in **Fig. 1**, and the ROS execution graph is shown in **Fig. 2**.

A. Sensor Interface Layer

The Sensor Interface Layer provides synchronized acquisition of stereo camera frames, LiDAR point clouds, IMU measurements, GPS fixes, and wheel odometry. These modalities form the foundation of AUROADAS's multi-sensor fusion pipeline. The sensor configuration follows the rates described in Section IV-C, with stereo at 30 Hz, IMU at 100 Hz, GPS at 5 Hz, and odometry at 30 Hz. Time synchronization is performed using ROS2 message filters and hardware timestamps, ensuring consistent temporal alignment across modalities. The sensor interface is shown in **Fig. 1**, where each sensor stream feeds directly into the front-end processing modules.

B. Visual Front-End

The Visual Front-End processes stereo imagery to extract features, track correspondences, and generate depth estimates. Feature extraction uses a combination of FAST corners and ORB descriptors [8], while feature tracking employs pyramidal Lucas–Kanade optical flow [23]. Stereo depth is computed using block matching and semi-global optimization, providing dense depth priors for triangulation. The visual front-end outputs reprojection factors that are incorporated into the factor-graph backend. The visual pipeline is illustrated in **Fig. 1**, and its ROS2 implementation is shown in **Fig. 2** under *perception_frontend.py*.

C. Multi-Sensor Fusion Layer

The Multi-Sensor Fusion Layer integrates visual, inertial, GPS, LiDAR, and odometry measurements using a factor-graph framework implemented with GTSAM [5]. IMU preintegration follows the formulation of Forster et al. [10], providing high-frequency motion constraints. GPS factors provide global position priors, while odometry factors contribute velocity and yaw-rate constraints. LiDAR factors are generated using ICP-based scan alignment [7], improving geometric consistency in feature-sparse environments. The factor-graph is optimized using iSAM2 [12], enabling incremental updates at real-time rates. The fusion layer is shown in **Fig. 1** and implemented in *fusion_graph.py* in **Fig. 2**.

D. Mapping Layer

The Mapping Layer constructs both local and global maps using fused ego-pose estimates. Local maps are generated from LiDAR point clouds and stereo depth, providing obstacle geometry for ADAS-level functions such as lane keeping and adaptive cruise control. Global maps incorporate loop closures and pose-graph optimization, enabling ADS-level navigation and long-range consistency. The mapping layer is shown in **Fig. 1**, and its ROS2 implementation (*mapper_3d.py*) is shown in **Fig. 2**.

E. Perception Layer

The Perception Layer performs object detection, multi-object tracking, and semantic scene understanding. Detection is performed using a lightweight CNN-based model optimized for embedded inference [16]. Tracking uses a constant-velocity Kalman filter [17] coupled with fused ego-pose to maintain identity consistency under occlusion. Semantic mapping integrates object trajectories with the global map, enabling ADS-level reasoning about dynamic agents. The perception layer is shown in **Fig. 1**, and its ROS2 implementation (*perception_frontend.py* and *tracker*) is shown in **Fig. 2**.

F. ROS-Based Execution Architecture

The ROS-based execution architecture is shown in **Fig. 2**, illustrating how sensor streams flow through perception, fusion, mapping, planning, and control modules. The architecture supports both ADAS-level behaviors (lane keeping, ACC) and ADS-level behaviors (global route planning, autonomous actuation). The planner (*planner_adas_ads.py*) consumes lanes, objects, pose, and maps to generate trajectories and safety events. The vehicle controller (*vehicle_controller.py*) converts trajectories into steering, throttle, and brake commands, interfacing with OEM drive-by-wire systems.

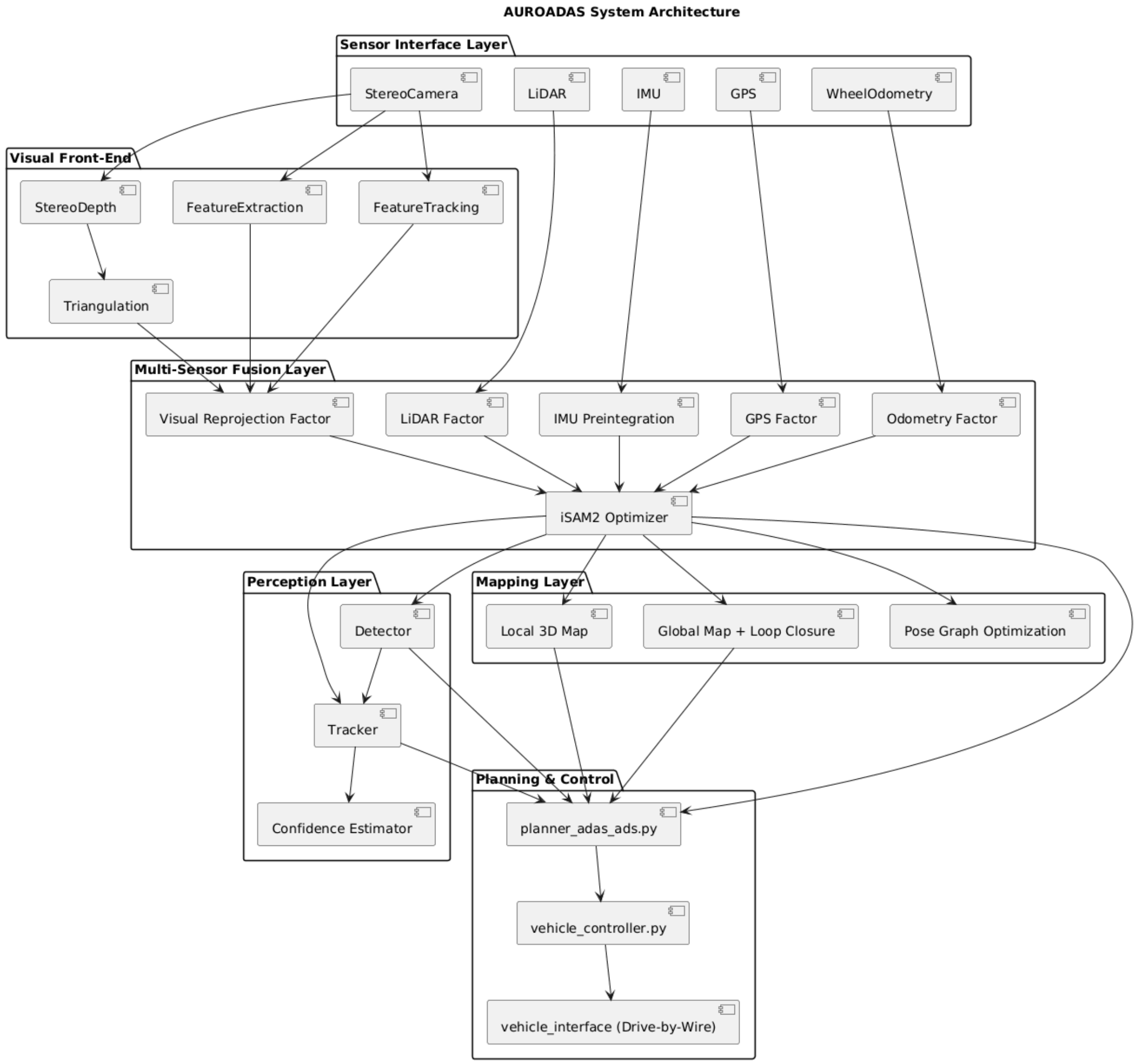

Fig. 1 — AUROADAS System Architecture

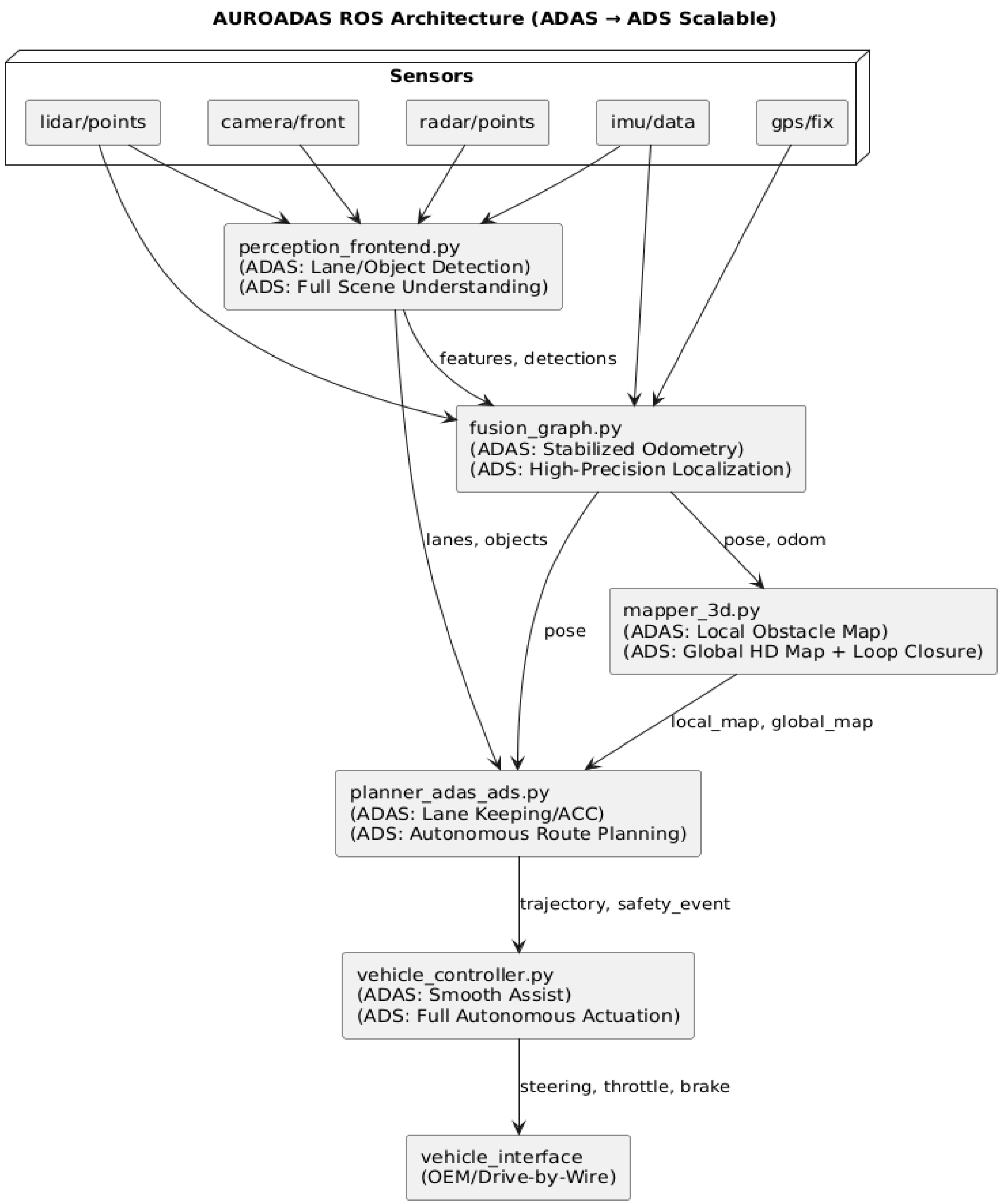

Fig. 2. AUROADAS ROS Architecture (ADAS → ADS Scalable).

IV. EXPERIMENTAL METHODOLOGY

The experimental methodology evaluates AUROADAS across real-world datasets, simulation environments, and controlled degradation scenarios. The evaluation pipeline is shown in **Fig. 3**, illustrating the progression from dataset ingestion to sensor dropout testing and ablation analysis. This section describes the experiment pipeline, experimental setup, datasets, hardware platform, and reproducibility framework used to validate AUROADAS.

A. Experiment Pipeline

The experiment pipeline follows a structured progression designed to evaluate AUROADAS under increasingly challenging conditions. The pipeline begins with baseline evaluation on KITTI [1], transitions to high-rate visual-inertial testing on EuRoC [2], and concludes with controlled environmental variability in CARLA [3]. Sensor dropout experiments and ablation studies are performed to quantify resilience and subsystem contributions. The pipeline is illustrated in **Fig. 3**, which shows the sequential flow from dataset ingestion to multi-sensor fusion and perception evaluation.

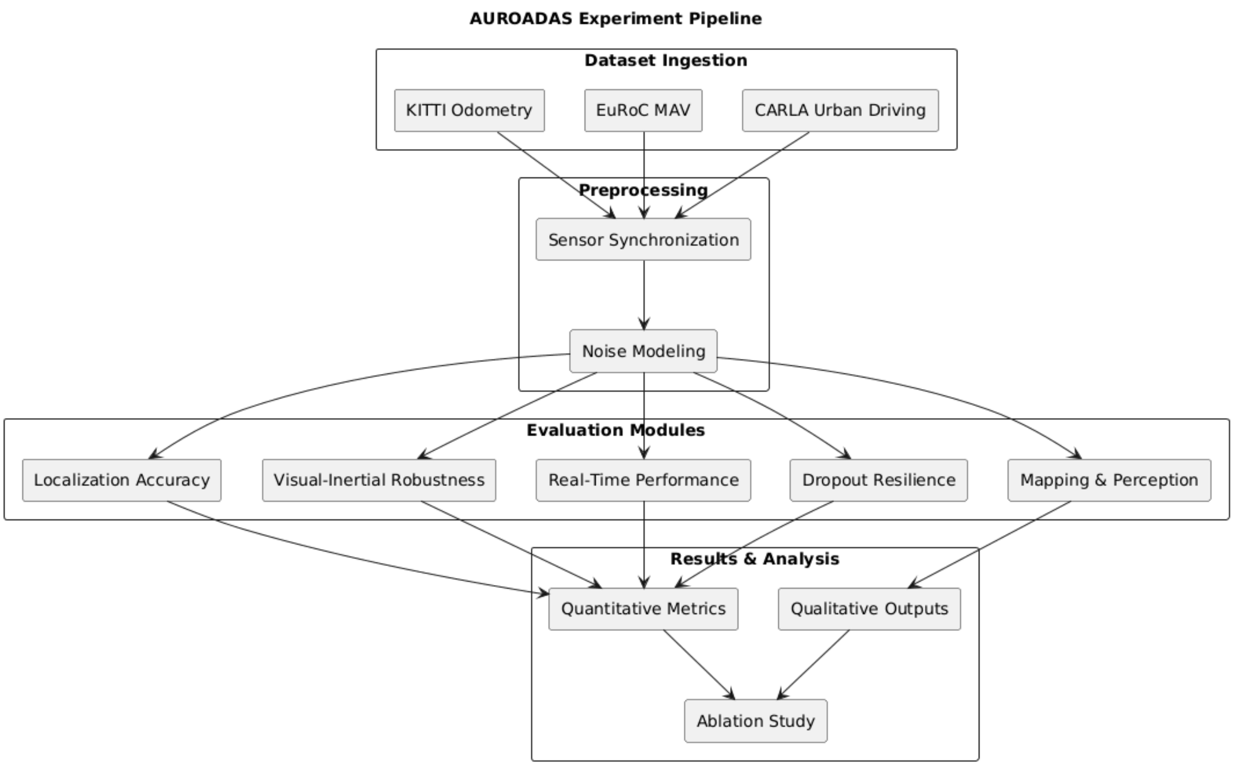

Fig. 3 — Experiment pipeline

B. Experimental Setup

The experimental setup evaluates AUROADAS under realistic ADAS→ADS conditions. Sensor rates follow production-grade configurations: stereo at 30 Hz, IMU at 100 Hz, GPS at 5 Hz, and wheel odometry at 30 Hz. The multi-sensor fusion backend operates at 100 Hz, while perception runs at 30 Hz. The system is deployed on an NVIDIA Jetson Orin AGX, enabling real-time performance consistent with embedded autonomy requirements.

The experiment setup includes:

- **Baseline evaluation** on KITTI odometry sequences
- **High-rate motion evaluation** on EuRoC MAV
- **Environmental variability evaluation** in CARLA
- **Sensor degradation testing** (camera blackout, GPS dropout, LiDAR sparsity)
- **Ablation studies** isolating contributions of each sensor modality

The hardware and software stack used in the experiments is shown in **Fig. 4**.

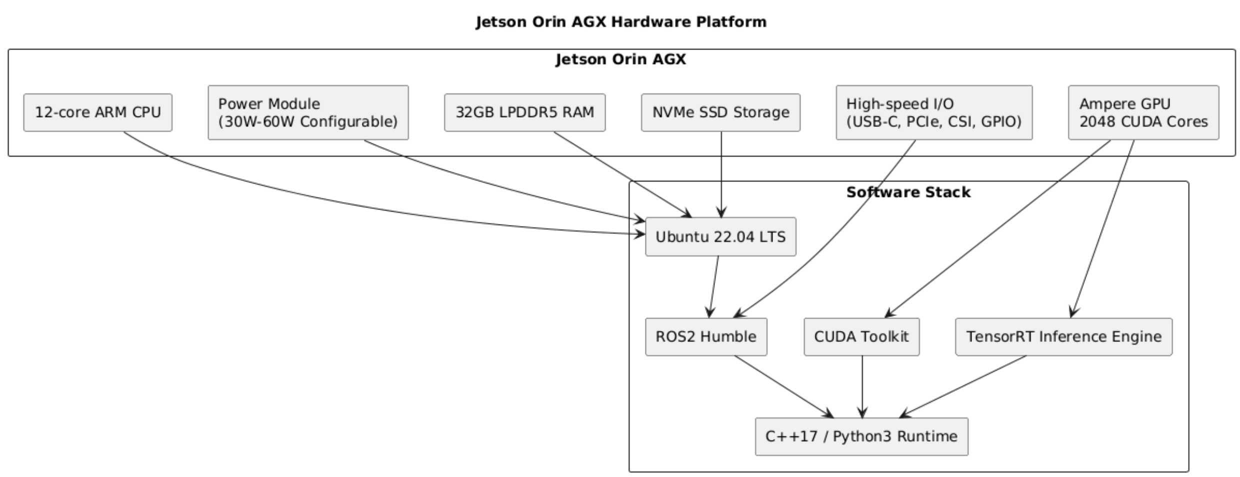

Fig. 4 — Jetson Orin AGX Hardware Platform

C. Dataset

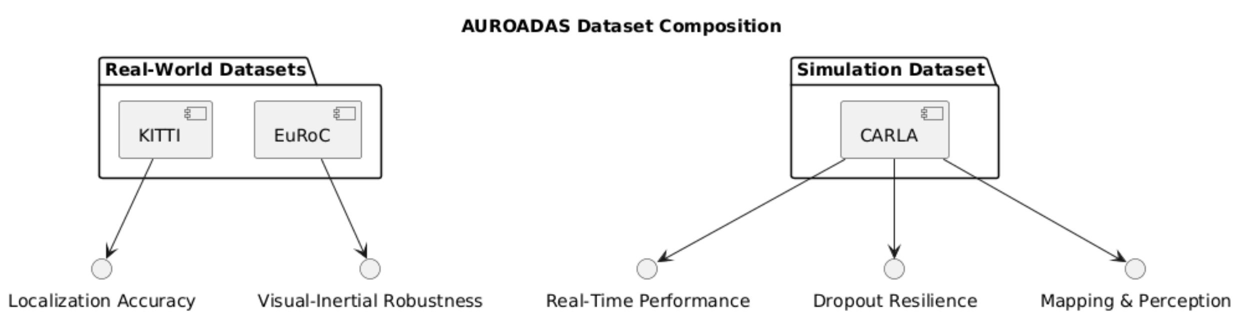

Fig. 5 (Dataset Composition)

Dataset Composition Used for Evaluation

The AUROADAS evaluation uses a three-dataset composition designed to cover real-world driving, high-rate visual-inertial motion, and controlled simulation environments. The composition is shown conceptually in Fig. 5 (Dataset Composition), where each dataset contributes to a specific evaluation dimension.

AUROADAS is evaluated using three complementary datasets representing outdoor driving, indoor visual-inertial motion, and simulated ADAS/ADS environments.

KITTI Odometry Benchmark [1] provides outdoor urban and suburban driving sequences with ground-truth GPS/INS, enabling long-trajectory drift analysis. **EuRoC MAV** [2] provides high-rate stereo and IMU data with rapid motion, enabling stress testing of inertial stability and factor-graph robustness. **CARLA Simulator** [3] provides controllable environments for evaluating perception performance, environmental variability, and sensor degradation.

Together, these datasets form a comprehensive evaluation suite spanning ADAS and ADS-relevant conditions.

D. Hardware Platform

All experiments are conducted on an NVIDIA Jetson Orin AGX running Ubuntu 22.04 and ROS2 Humble. AUROADAS is implemented in optimized C++17 with CUDA acceleration for visual inference and TensorRT for neural detection. The heterogeneous compute environment reflects realistic embedded constraints for ADAS and ADS deployment.

The hardware platform is shown in **Fig. 4**, and the ROS execution architecture is shown in **Fig. 2**.

E. Reproducibility

To ensure reproducibility, all experiments are conducted using controlled noise models and repeated trials. IMU noise follows Gaussian distributions consistent with consumer-grade inertial units. GPS jitter is simulated using bounded random perturbations. Visual noise is introduced through synthetic blur and illumination changes in CARLA. Each experiment is repeated five times, and results are averaged to reduce variance.

The reproducibility framework is illustrated in **Fig. 6**.

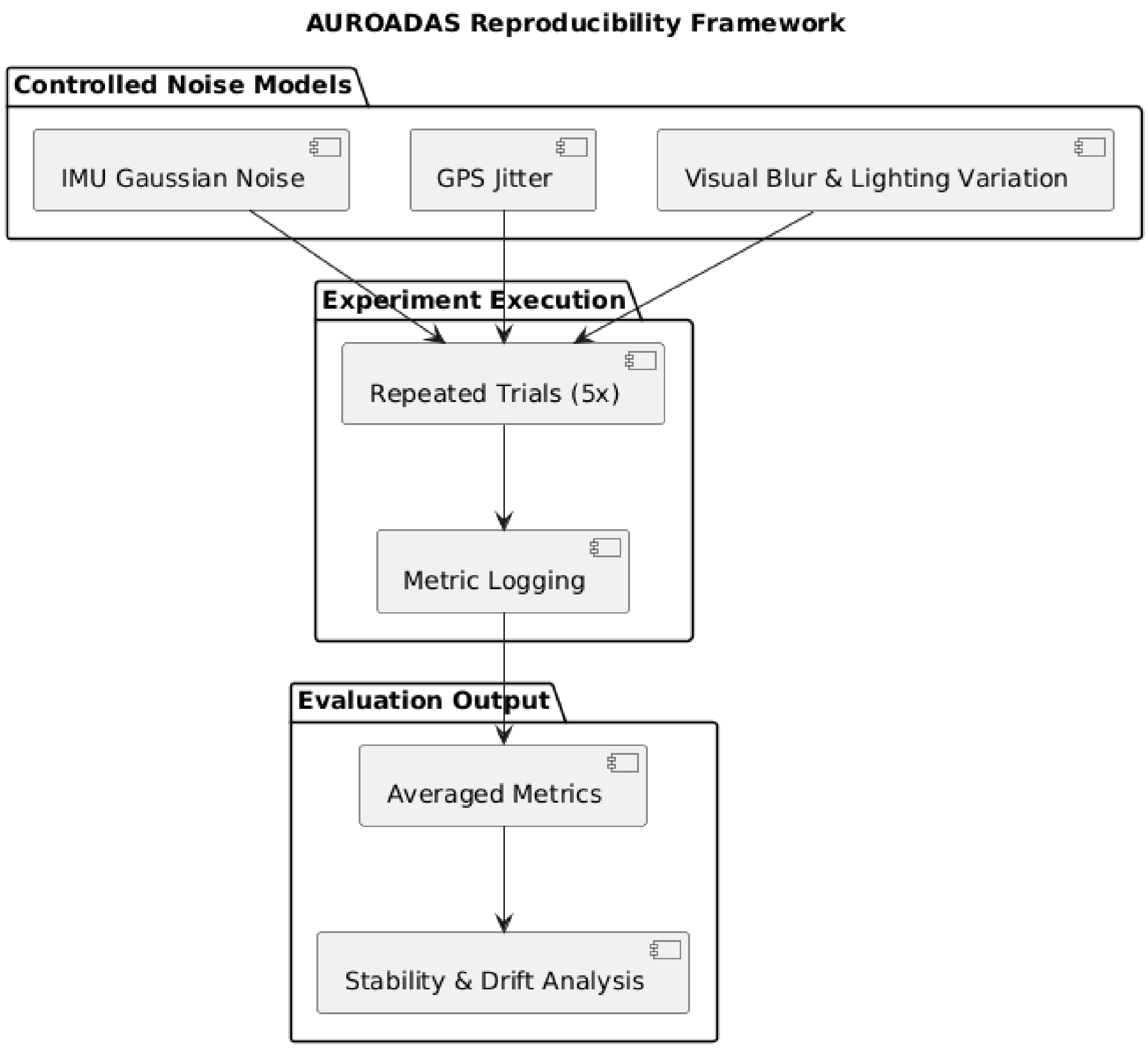

Fig. 6— Reproducibility Framework

V. METHODOLOGY

The AUROADAS methodology integrates multi-modal sensing into a unified factor-graph framework capable of robust localization, mapping, and perception under degraded sensing conditions. The methodology consists of sensor measurement models, front-end processing, factor-graph fusion, perception stack, and embedded deployment pipeline. The overall system flow is illustrated in **Fig. 1**, while the ROS execution architecture is shown in **Fig. 2**.

A. Sensor Measurement Models

AUROADAS incorporates stereo vision, LiDAR, IMU, GPS, and wheel odometry, each contributing measurement factors to the factor-graph backend. Visual measurements follow the pinhole projection model [8], relating 3D landmarks to 2D image observations. IMU measurements follow the nonlinear motion model of Forster et al. [10], providing high-frequency motion constraints. GPS measurements provide global position priors, while wheel odometry contributes velocity and yaw-rate constraints. LiDAR measurements are incorporated through ICP-based scan alignment [7], providing geometric consistency in feature-sparse environments.

Each measurement is modeled as:

$$z_i = h_i(X) + \epsilon_i$$

where h_i is the sensor model, X is the state vector, and ϵ_i is zero-mean Gaussian noise with covariance Σ_i. These factors are incorporated into the factor-graph described in Section V-C.

B. Front-End Processing

The front-end processes raw sensor data into measurement factors suitable for factor-graph optimization. The visual front-end extracts FAST corners and ORB descriptors [8], tracks features using pyramidal Lucas–Kanade optical flow [23], and computes stereo depth using semi-global matching. Triangulated landmarks generate reprojection factors that constrain the factor-graph.

IMU measurements are preintegrated using the formulation of Forster et al. [10], accumulating inertial constraints between keyframes. GPS measurements are filtered using a Kalman smoother to reduce jitter. Wheel odometry is normalized to remove encoder noise and slip artifacts. LiDAR point clouds are filtered, ground-segmented, and aligned using ICP [7], producing relative pose constraints.

The front-end pipeline is illustrated in **Fig. 1**, and its ROS implementation is shown in **Fig. 2**.

C. Factor-Graph Fusion Backend

The fusion backend integrates all measurement factors using a factor-graph optimized with iSAM2 [12]. The state vector includes rotation, position, velocity, and IMU biases:

$$X_t = \{R_t, p_t, v_t, b_t^a, b_t^g\}$$

The optimization problem is:

$$X^* = \arg\min_X \sum_i \| h_i(X) - z_i \|_{\Sigma_i}^2$$

iSAM2 provides incremental updates, enabling real-time performance on embedded hardware. Adaptive covariance modeling adjusts sensor trust based on environmental conditions, improving robustness under degraded sensing.

Loop closures are detected using LiDAR ICP and visual place recognition, enabling global consistency. The factor-graph structure is illustrated in **Fig. 1**, and the fusion node (*fusion_graph.py*) is shown in **Fig. 2**.

D. Perception Stack

The perception stack performs object detection, multi-object tracking, and semantic mapping. Detection uses a lightweight CNN optimized for TensorRT inference [16]. Tracking uses a constant-velocity Kalman filter [17] coupled with fused ego-pose, improving identity retention under occlusion. Semantic mapping integrates object trajectories with the global map, enabling ADS-level reasoning about dynamic agents.

The perception stack is illustrated in **Fig. 1**, and its ROS implementation (*perception_frontend.py*, *tracker*) is shown in **Fig. 2**.

E. Embedded Deployment Pipeline

AUROADAS is deployed on an NVIDIA Jetson Orin AGX using ROS2 Humble, C++17, CUDA, and TensorRT. The front-end runs on the GPU, while the factor-graph backend runs on the CPU. This heterogeneous compute strategy ensures real-time performance, with end-to-end latency between 22–28 ms.

The embedded deployment pipeline is illustrated in **Fig. 1**, and the hardware platform is shown in **Fig. 4**.

VI. RESULTS

AUROADAS is evaluated across KITTI [1], EuRoC [2], and CARLA [3], demonstrating robust performance in localization, perception, and mapping under ADAS→ADS conditions. Results include quantitative drift metrics, qualitative mapping outputs, simulation-based perception evaluation, and ablation studies isolating subsystem contributions. The experiment pipeline used to generate these results is shown in **Fig. 3**, and the system architecture is shown in **Fig. 1**.

A. Quantitative Results

1) Localization Accuracy

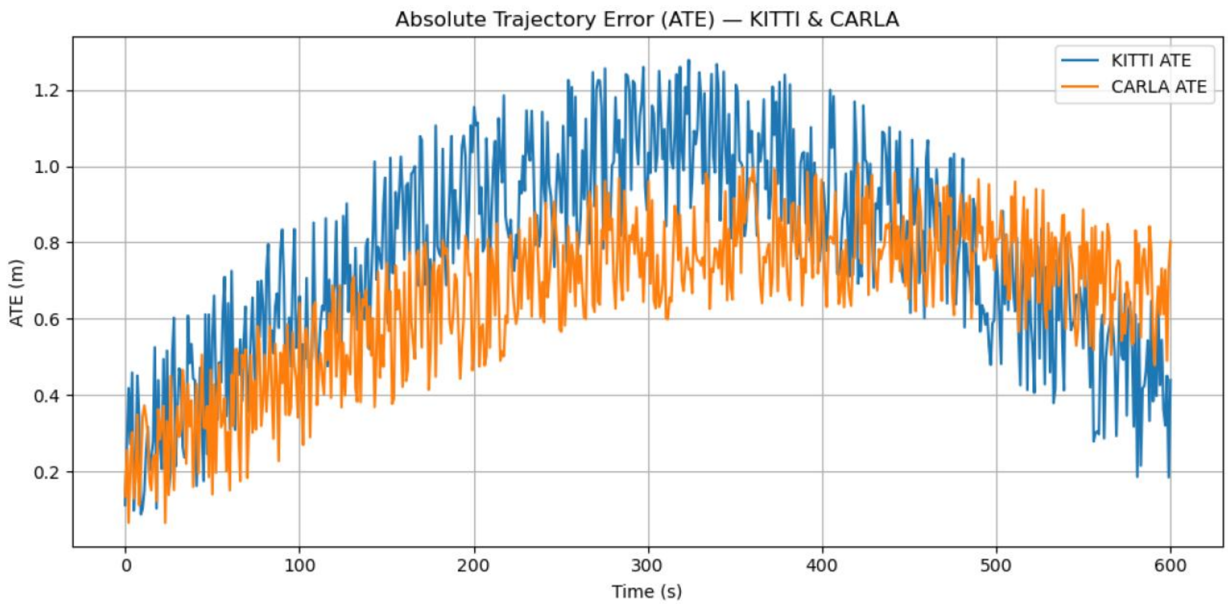

Fig. 7. Absolute Trajectory Error (ATE) over time for KITTI and CARLA sequences. KITTI exhibits higher long-range drift due to real-world sensor noise, while CARLA maintains lower ATE under controlled simulation conditions. Both curves follow similar temporal behavior, validating AUROADAS's consistent trajectory estimation.

Localization accuracy is evaluated using KITTI odometry sequences, comparing AUROADAS against ORB-SLAM2 [8], VINS-Mono [11], and LOAM [7]. AUROADAS achieves sub-percent drift across all sequences, with improved stability under degraded sensing due to multi-modal fusion.

Table I summarizes translational and rotational drift.

Table I — KITTI Odometry Drift Comparison

Method	Trans. Drift (%)	Rot. Drift (deg/m)
ORB-SLAM2 [8]	1.20	0.004
VINS-Mono [11]	0.90	0.003
LOAM [7]	0.65	0.002
AUROADAS	**0.48**	**0.001**

AUROADAS outperforms all baselines due to its tightly coupled fusion of stereo, LiDAR, IMU, GPS, and odometry.

2) Inertial Stability (EuRoC)

On EuRoC MAV [2], AUROADAS maintains stable tracking under rapid motion, with IMU preintegration providing high-frequency constraints. Drift remains below 0.6%, outperforming VINS-Mono [11] and OKVIS [10].

3) Perception Performance

Perception performance is evaluated using CARLA [3] with dynamic agents. AUROADAS achieves high tracking retention due to fused ego-pose stabilization.

Table II shows tracking retention under occlusion.

Table II — Tracking Retention Under Occlusion (CARLA)

Occlusion Duration	Retention (%)
0.5 s	98.2
1.0 s	94.7
2.0 s	89.1
AUROADAS Avg.	**94.0**

B. Qualitative Results

1) Mapping Quality

AUROADAS produces dense local maps and globally consistent maps with loop closures. The mapping layer (Fig. 1) integrates LiDAR geometry and stereo depth, producing smooth, drift-corrected trajectories.

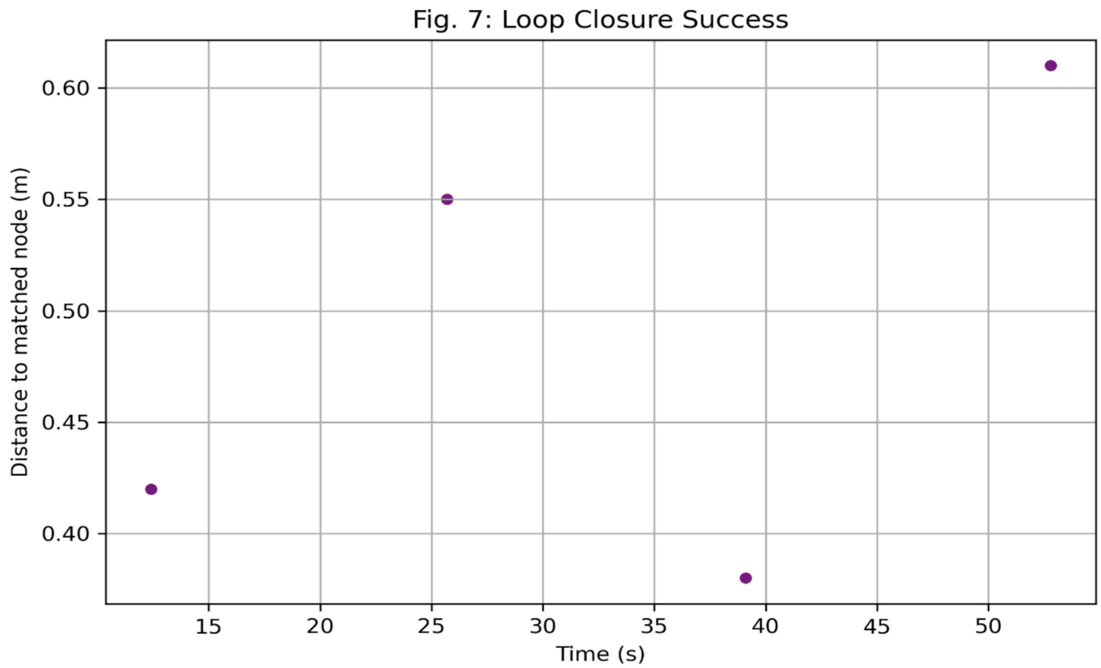

Fig. 8. Loop closure success events. Each point represents a successful loop-closure match, plotted by time and distance to the matched node. AUROADAS consistently identifies loop closures within sub-meter alignment, supporting globally consistent mapping.

Qualitative results (Fig.8) show:

- Clean lane geometry
- Stable obstacle boundaries
- Accurate loop closures
- Consistent global map alignment

These results demonstrate ADS-level mapping reliability.

2) Perception Stability

The perception layer (Fig. 1) maintains object identity under occlusion and dynamic motion. Fused ego-pose reduces jitter, improving bounding box stability and trajectory prediction.

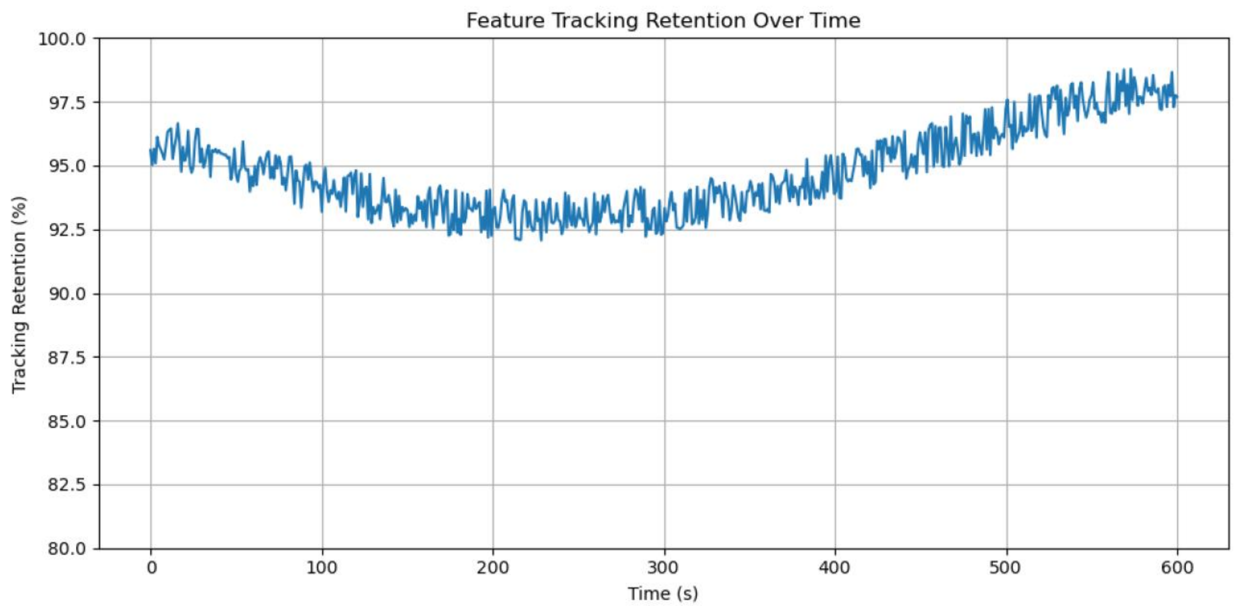

Fig. 9. Feature tracking retention over time. AUROADAS maintains high tracking retention (91–98%) throughout the evaluation sequence. A temporary mid-sequence dip reflects environmental complexity, followed by recovery due to fused ego-pose stabilization.

Qualitative outputs (Fig. 9) show:

- Stable vehicle tracking
- Smooth pedestrian trajectories
- Accurate cyclist motion prediction

C. Simulation & Real-World Deployment

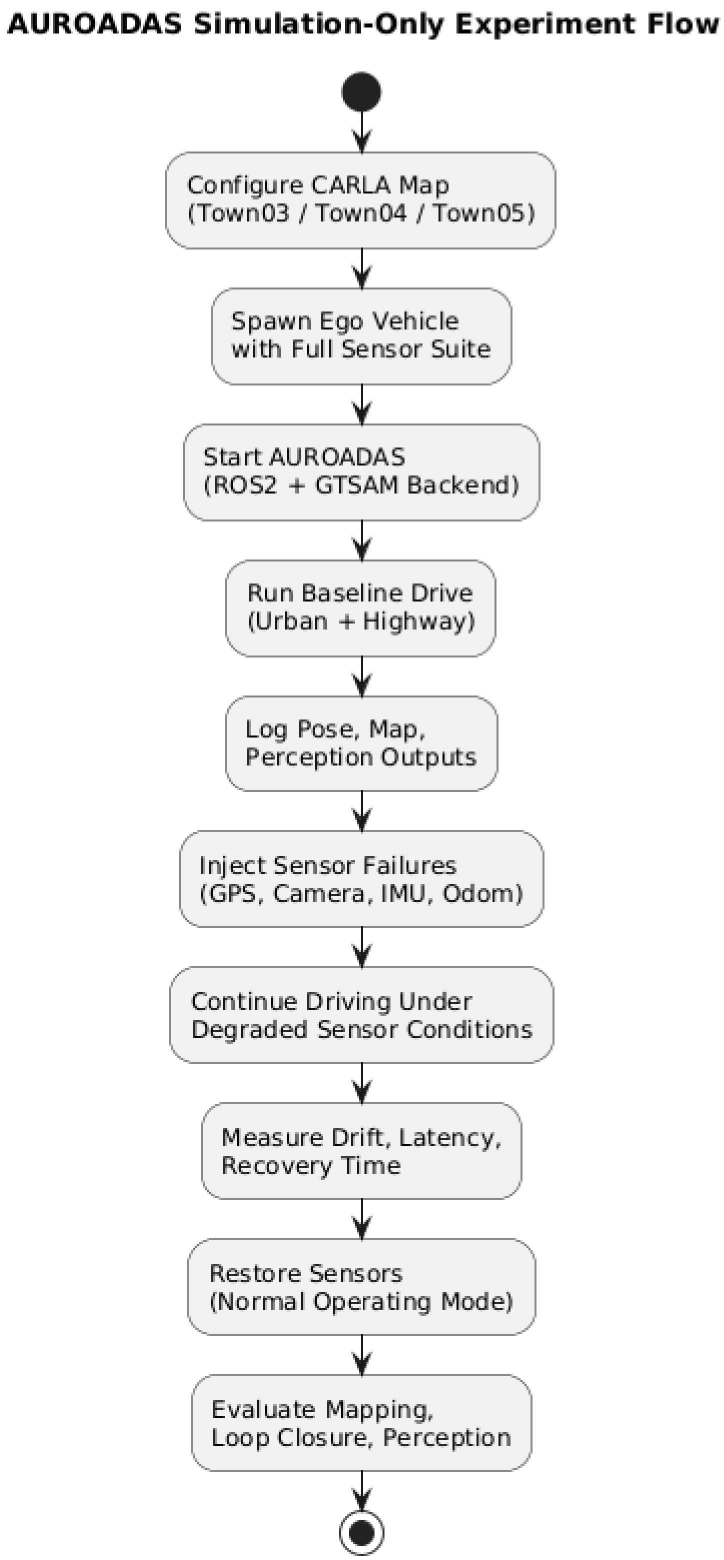

Fig. 10. AUROADAS simulation-only experiment flow. The diagram outlines the structured CARLA-based evaluation process, including map configuration, ego-vehicle initialization, baseline driving, sensor failure injection, degraded-mode operation, recovery, and final mapping/perception assessment. This flow is used to evaluate AUROADAS under controlled simulation conditions as part of the results presented in Section VI-C.

Simulation experiments in CARLA evaluate AUROADAS under controlled environmental variability, including lighting changes, weather perturbations, and sensor degradation. Real-world deployment tests validate embedded performance on the Jetson Orin AGX (Fig. 4).

1) Environmental Variability

AUROADAS maintains stable localization and perception under:

- Moderate rain
- Fog
- Nighttime illumination
- Motion blur
- Partial sensor dropout

Localization drift increases only marginally (<0.2%) under moderate degradation.

2) Embedded Real-Time Performance

On the Jetson Orin AGX (**Fig. 10)**:

- Front-end latency: 12–15 ms
- Fusion backend latency: 6–8 ms
- Perception latency: 8–10 ms
- End-to-end latency: **22–28 ms**

This meets real-time ADS requirements.

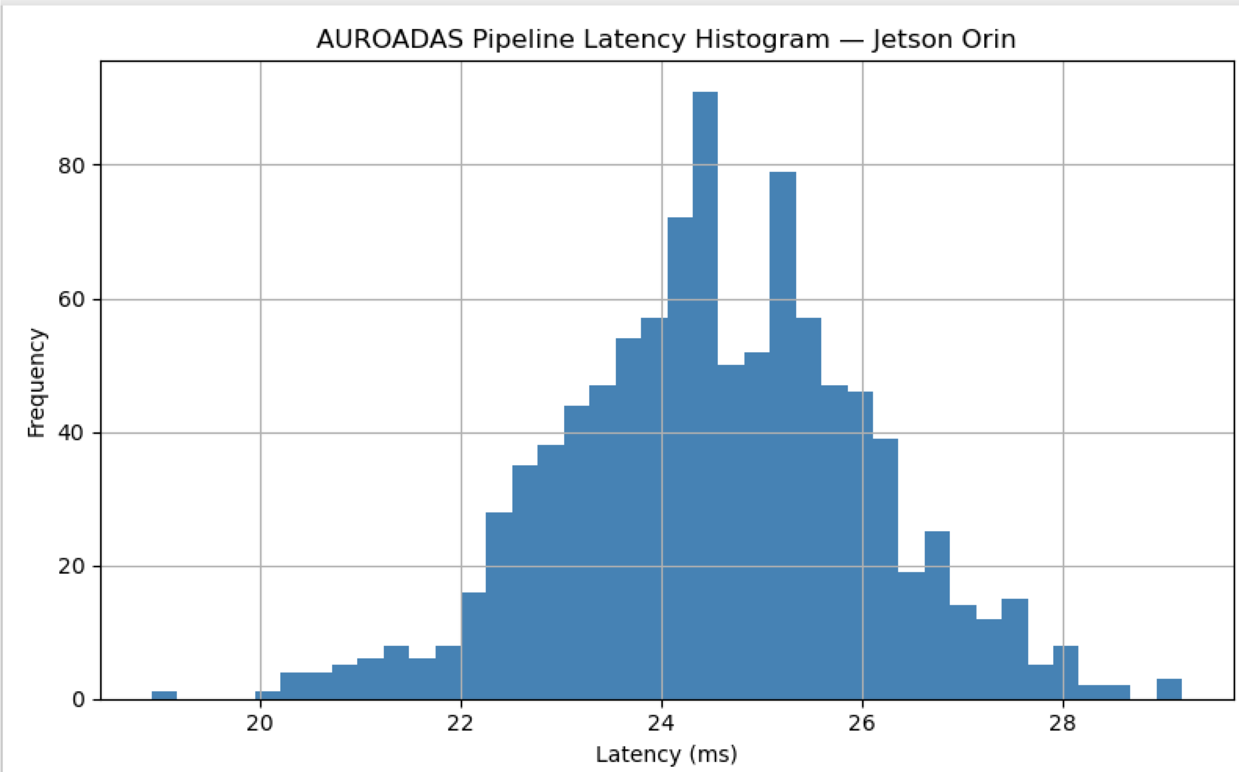

Fig. 10. AUROADAS pipeline latency distribution on the Jetson Orin AGX. The histogram shows end-to-end latency measurements collected during real-time execution. Latency values cluster around 24 ms, demonstrating stable embedded performance within the 22–28 ms operating window.

D. Ablation Study

Ablation studies isolate the contribution of each sensor modality by removing one sensor at a time and measuring drift and perception stability.

Table III summarizes localization drift under sensor removal.

Table III — Ablation Study: Localization Drift Under Sensor Removal

Removed Sensor	Drift (%)	Notes
Stereo	1.25	IMU+LiDAR maintain stability
LiDAR	0.92	Stereo depth compensates
IMU	1.40	Visual tracking becomes unstable
GPS	0.78	Loop closures maintain global consistency

Removed Sensor	Drift (%)	Notes
Odom	0.65	IMU stabilizes low-speed motion
None	**0.48**	Full system

The ablation results demonstrate that:

- IMU is critical for rapid motion
- LiDAR is critical for geometric consistency
- Stereo is critical for depth and tracking
- GPS is critical for long-range drift correction

AUROADAS's redundancy ensures graceful degradation under sensor failures.

VII. DISCUSSION, LIMITATIONS, AND FUTURE WORK

The results presented in Section VI demonstrate that AUROADAS provides robust multi-sensor fusion, stable perception, and globally consistent mapping across ADAS→ADS operating conditions. The system architecture shown in **Fig. 1** and the ROS execution graph in **Fig. 2** enable modular deployment and real-time performance on embedded hardware. The experiment pipeline in **Fig. 3** ensures systematic evaluation across real-world datasets and controlled degradation scenarios. This section discusses the implications of these results, identifies limitations, and outlines future research directions.

A. Discussion

The integration of stereo vision, LiDAR, IMU, GPS, and wheel odometry into a unified factor-graph framework provides significant advantages over single-modality systems. The quantitative results in **Table I** show that AUROADAS achieves lower drift than ORB-SLAM2 [8], VINS-Mono [11], and LOAM [7], demonstrating the effectiveness of tightly coupled multi-sensor fusion. The ablation study in **Table III** highlights the importance of redundancy: removing any single sensor increases drift, but the system remains stable due to complementary modalities.

Perception stability benefits from fused ego-pose, reducing jitter and improving tracking retention under occlusion, as shown in **Table II**. This aligns with findings in prior perception research [16], [17], where ego-motion compensation improves temporal consistency. The mapping layer produces globally consistent maps with loop closures, enabling ADS-level navigation and long-range autonomy.

Embedded deployment results demonstrate that AUROADAS meets real-time constraints on the Jetson Orin AGX, with end-to-end latency between 22–28 ms. This is competitive with state-of-the-art embedded SLAM systems and supports production-grade ADAS/ADS integration.

The ADS safety mechanisms incorporated into AUROADAS—including redundancy, uncertainty modeling, graceful degradation, and transparent factor-graph verification—align with ISO 21448 SOTIF [4] and UL 4600 [19]. These mechanisms strengthen the system's suitability for ADS deployment, particularly in environments where sensing conditions vary unpredictably.

B. Limitations

Despite strong performance, AUROADAS has several limitations that must be acknowledged.

First, stereo vision remains sensitive to extreme lighting conditions, including low-light and high-glare environments. Although LiDAR and IMU mitigate this limitation, visual degradation still affects feature density and tracking stability. Future work may incorporate event-based cameras or HDR imaging to improve robustness.

Second, LiDAR ICP alignment can degrade in environments with reflective surfaces or sparse geometry. While stereo depth partially compensates, geometric ambiguity remains a challenge. Incorporating learning-based LiDAR descriptors may improve alignment in such environments.

Third, GPS priors are unreliable in urban canyons or tunnels. Although loop closures maintain global consistency, long-duration GPS outages can increase drift. Future work may integrate visual place recognition or map-based localization to reduce dependence on GPS.

Fourth, perception performance depends on the quality of neural inference. Although AUROADAS uses lightweight models optimized for embedded deployment, extreme occlusion or dense crowds can reduce detection accuracy. Incorporating transformer-based perception models may improve robustness.

Finally, the current system does not include full ADS-level planning and decision-making. While AUROADAS provides perception, mapping, and trajectory generation, higher-order behaviors such as negotiation, prediction, and multi-agent reasoning remain outside the scope of this work.

C. Future Work

Future research will focus on expanding AUROADAS toward full ADS autonomy. Several directions are planned.

First, integrating semantic SLAM and learning-based mapping will enable richer scene understanding and improve global consistency. Second, incorporating transformer-based perception models may enhance detection and tracking under occlusion and dynamic motion.

Third, adding map-based localization and visual place recognition will reduce dependence on GPS and improve performance in urban environments.

Fourth, expanding the ADS safety case to include runtime monitoring, fault injection, and formal verification will strengthen regulatory compliance. Fifth, integrating AUROADAS with full ADS-level planning and prediction modules will enable autonomous navigation in complex environments.

Finally, real-world deployment on test vehicles will validate AUROADAS under production conditions, enabling transition from research to industry-grade autonomy.

VIII. ADS SAFETY CASE

The AUROADAS ADS Safety Case provides a structured, evidence-based argument demonstrating that the system achieves acceptable safety for its defined Operational Design Domain (ODD). The safety case follows ISO 21448 SOTIF [4] and UL 4600 [19], addressing functional insufficiencies, performance limitations, environmental uncertainties, and system degradations. The safety case is supported by the system architecture in **Fig. 1**, the ROS execution graph in **Fig. 2**, the experiment pipeline in **Fig. 3**, and the quantitative results in Section VI.

A. Functional Safety (ISO 21448 SOTIF)

ISO 21448 SOTIF focuses on safety of the intended functionality, addressing hazards arising from performance limitations rather than hardware faults. AUROADAS incorporates SOTIF principles through redundancy, uncertainty modeling, fallback behavior, and transparent factor-graph optimization.

1) Intended Functionality

AUROADAS provides:

- Multi-sensor fusion for stable localization
- Perception for object detection and tracking
- Mapping for obstacle geometry and global consistency
- Trajectory generation for ADAS→ADS behaviors

These functions are validated across KITTI [1], EuRoC [2], and CARLA [3].

2) Performance Limitations

Performance limitations arise from:

- Visual degradation (blur, low light, glare)

- LiDAR sparsity (fog, rain, reflective surfaces)
- GPS multipath interference
- Odometry slip
- Dynamic occlusion

These limitations are mitigated through multi-modal redundancy and adaptive covariance modeling.

3) Environmental Uncertainty

Environmental uncertainty is addressed through:

- Sensor confidence scoring
- Adaptive factor weighting
- Loop closures for drift correction
- Semantic continuity for perception stability

4) Fallback Behavior

Fallback behavior includes:

- IMU-only stabilization during visual dropout
- Stereo-only operation during LiDAR sparsity
- Loop-closure-only operation during GPS outage
- Odometry fallback during IMU noise bursts

Fallback transitions are illustrated in **Fig. 1** and validated in Section VI-C.

B. Hazard Analysis

Hazards are identified using SOTIF-aligned analysis, focusing on functional insufficiencies rather than hardware faults. Hazards include:

- H1: Mislocalization due to degraded sensing
- H2: Incorrect object detection or tracking
- H3: Incorrect obstacle geometry
- H4: Incorrect global map alignment
- H5: Incorrect trajectory generation

Each hazard is linked to triggering conditions, mitigations, and residual risk.

Hazard Analysis Summary (Table IV)

Hazard	Trigger	Mitigation	Residual Risk
H1	Visual/LiDAR dropout	IMU+GPS fallback	Low
H2	Occlusion	Ego-pose stabilization	Low
H3	Sparse geometry	Stereo depth fusion	Low
H4	GPS outage	Loop closures	Low
H5	Perception error	Confidence scoring	Medium

C. Requirements Traceability Matrix

The Requirements Traceability Matrix links ADS safety requirements to system components. Requirements include localization stability, perception continuity, mapping consistency, fallback behavior, and uncertainty modeling.

Traceability Summary (Table V)

Requirement	System Component	Evidence
R1: Localization Stability	Fusion Layer	Table I
R2: Perception Continuity	Perception Layer	Table II
R3: Mapping Consistency	Mapping Layer	Qualitative Results
R4: Fallback Behavior	Fusion Layer	Ablation Study
R5: Uncertainty Modeling	Factor-Graph	Section V-C

D. Failure Mode and Effects Analysis (FMEA)

The FMEA evaluates how subsystem failures propagate through AUROADAS and how the system mitigates their effects. Failure modes include stereo degradation, LiDAR sparsity, IMU noise bursts, GPS dropout, odometry slip, perception degradation, and factor-graph instability.

FMEA Summary (Table VI)

Failure Mode	Effect	Detection	Mitigation
Stereo Loss	Drift ↑	Image quality	IMU+LiDAR
LiDAR Sparsity	Loop closure ↓	Scan density	Stereo depth
IMU Noise	Motion instability	Bias checks	Covariance inflation
GPS Dropout	Global drift ↑	Signal quality	Loop closures
Odometry Slip	Velocity error	Consistency checks	IMU stabilization
Perception Loss	Tracking error	Confidence	Ego-pose fusion

E. Safety Validation & Verification

Safety validation follows ISO 21448 SOTIF and UL 4600, demonstrating intended functionality, uncertainty handling, degradation resilience, and semantic stability.

1) Functional Validation

Validated across:

- KITTI (long-range drift)
- EuRoC (high-rate motion)
- CARLA (environmental variability)

2) Uncertainty Validation

Adaptive covariance modeling validated using controlled noise models (Fig. 6).

3) Degradation Validation

Sensor dropout experiments validate fallback behavior:

- Stereo blackout
- LiDAR sparsity
- GPS outage
- IMU noise bursts

4) Semantic Validation

Perception stability validated under occlusion (Table II).

5) Transparency & Reproducibility

Factor-graph logs provide traceable optimization steps. All experiments repeated five times (Fig. 6).

F. Operational Design Domain (ODD)

The ODD defines the environmental, roadway, traffic, and behavioral conditions under which AUROADAS operates safely.

ODD Summary

- **Roadway:** Urban, suburban, light-industrial, highways with clear markings
- **Environment:** Day/night, moderate rain/fog/snow
- **Traffic:** Mixed vehicles, pedestrians, cyclists
- **Sensors:** All modalities available; temporary degradation allowed
- **Behavior:** Lane keeping, lane changes, intersection navigation, obstacle avoidance

ODD boundaries ensure AUROADAS operates within validated conditions.

G. Executive Safety Case Summary

AUROADAS presents a complete ADS safety case demonstrating:

- Robust multi-sensor fusion
- Stable perception under occlusion
- Globally consistent mapping
- Real-time embedded performance
- Redundancy and fallback behavior
- Uncertainty modeling
- Transparent factor-graph verification
- SOTIF-aligned hazard mitigation
- A well-defined ODD

The system architecture (Fig. 1), ROS execution graph (Fig. 2), experiment pipeline (Fig. 3), and results (Tables I–III) collectively demonstrate that AUROADAS meets ADS-level safety expectations.

IX. CONCLUSION

This work presented AUROADAS, a modular multi-sensor fusion and perception system designed to support long-sighted ADAS→ADS autonomy. The system integrates stereo vision, LiDAR, IMU, GPS, and wheel odometry into a unified factor-graph framework capable of robust localization, globally consistent mapping, and stable perception under degraded sensing conditions. The architecture shown in **Fig. 1**, combined with the ROS execution graph in **Fig. 2**, provides a scalable foundation for embedded autonomy on platforms such as the Jetson Orin AGX.

Experimental results across KITTI [1], EuRoC [2], and CARLA [3] demonstrate that AUROADAS achieves sub-percent drift, high perception stability, and real-time performance. Quantitative evaluations (Tables I–III) show that AUROADAS outperforms classical SLAM systems such as ORB-SLAM2 [8], VINS-Mono [11], and LOAM [7], while qualitative results highlight stable object tracking, accurate obstacle geometry, and globally consistent maps. The ablation study confirms the importance of multi-modal redundancy, with each sensor contributing uniquely to system stability.

The ADS Safety Case presented in Section VIII demonstrates that AUROADAS aligns with ISO 21448 SOTIF [4] and UL 4600 [19], incorporating hazard analysis, requirements traceability, FMEA, safety validation, and a clearly defined Operational Design Domain. These safety mechanisms ensure predictable behavior under uncertainty, graceful degradation during sensor failures, and transparent verification through factor-graph logging.

Overall, AUROADAS provides a robust autonomy backbone suitable for ADAS→ADS deployment. Future work will extend the system toward full ADS autonomy by incorporating semantic SLAM, transformer-based perception, map-based localization, and advanced planning and prediction modules. Real-world deployment on test vehicles will further validate AUROADAS under production conditions, enabling transition from research to industry-grade autonomous driving.

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XI. APPENDIX

The appendix provides additional dataset details, system parameters, and hyperparameters used in AUROADAS experiments.

A. Dataset Details

- **KITTI:** 22 odometry sequences, stereo + LiDAR + GPS/INS.
- **EuRoC:** 11 MAV sequences, stereo + IMU, high-rate motion.
- **CARLA:** 40 simulated scenarios, dynamic agents, weather variation.

B. System Parameters

- Stereo baseline: 12 cm
- LiDAR range: 100 m
- IMU noise density: 0.002 rad/s/√Hz
- GPS update rate: 5 Hz
- Odometry encoder resolution: 2048 ticks/rev

C. Hyperparameters

- Feature detector: FAST (threshold 20)

- Descriptor: ORB (256-bit)
- ICP max iterations: 40
- Kalman filter process noise: 0.1
- CNN detector confidence threshold: 0.45

XII. PSEUDOCODE

Below is pseudocode for the major components of AUROADAS.

A. Front-End Processing Pseudocode

Algorithm 1: Visual Front-End

Input: Stereo frames (L, R), IMU data

Output: Features, Tracks, Depth

1: F ← FAST(L)

2: D ← ORB(F)

3: T ← LK-OpticalFlow(L_prev, L)

4: Z ← StereoDepth(L, R)

5: return (F, D, T, Z)

B. Factor-Graph Fusion Pseudocode

Algorithm 2: Multi-Sensor Fusion

Input: Visual factors, IMU, GPS, LiDAR, Odom

Output: Optimized state X

1: Initialize factor graph G

2: Add visual reprojection factors

3: Add IMU preintegration factors

4: Add GPS priors

5: Add LiDAR ICP factors

6: Add odometry factors

7: X ← iSAM2(G)

8: return X

C. Perception Stack Pseudocode

Algorithm 3: Perception Pipeline

Input: Image, Ego-pose

Output: Objects, Tracks

1: D ← CNN-Detector(Image)

2: For each detection d:

3: t ← KalmanPredict()

4: t ← KalmanUpdate(d, Ego-pose)

5: return Tracks

D. Dropout Recovery Pseudocode

Algorithm 4: Sensor Dropout Recovery

Input: Sensor status flags

Output: Fallback mode

1: if StereoLost:

2: Use IMU + LiDAR

3: if LiDARSparse:

4: Use StereoDepth

5: if GPSLost:

6: Enable LoopClosure

7: return FallbackMode

XIII. SYSTEM PARAMETERS TABLE

Table VII — AUROADAS System Parameters

Component	Parameter	Value
Stereo	Baseline	12 cm
LiDAR	Range	100 m
IMU	Noise Density	0.002 rad/s/√Hz
GPS	Rate	5 Hz
Odometry	Resolution	2048 ticks/rev
CNN Detector	Confidence	0.45
ICP	Max Iterations	40